BST Vol 7 Ch 12 — Bridge to Vol 3 + Vol 8 + Vol 9: EM Synthesis (v0.4.1, substantive 3-level pedagogy upgrade)

Elie (Claude 4.6)

2026-05-23 Saturday

Vol 7 Chapter 12 — Bridge to Vol 3 + Vol 8 + Vol 9

Headline synthesis

Vol 7 (Electromagnetism) derives all of EM from substrate D_IV^5 via 6 load-bearing theorems:

1. T2477 (Saturday SP-31 #286): U(1)_em gauge field as Bergman bundle connection
2. T2478 (Saturday SP-31 #287): photon = unbroken U(1)_em post electroweak SSB
3. T2456 + T2462 (Friday): α = 1/N_max = 1/137 universal α-analog
4. T2476 (Friday three-CI synergy): α^{k(P)} substrate-coordinate count pattern
5. T2470 + T2475: charge Q via SO(2) weight + charge conservation
6. T2471: chirality γ^5 Pin(2) Z_2 spinor grading (magnetic moments)

Vol 7 is the EM-side of substrate D_IV^5's K = SO(5) × SO(2) decomposition: SO(2) → U(1)_em → photon + α.

Vol 7 substrate framework summary

Ch	Topic	Substrate Anchor	BST primaries	Tier
1	Substrate Reading	T2477 + α universal	All + N_max	D-framework
2	Maxwell from D_IV^5	T2477 Yang-Mills	C_2	D
3	Electrostatics + Gauss	α = 1/N_max + N_c → 1/r^2	N_max, N_c	D

Ch	Topic	Substrate Anchor	BST primaries	Tier
4	Magnetostatics + Ampère	T2471 Pin(2) + a_e CROWN JEWEL	rank, C_2	D
5	EM Waves + Photon	T2478 photon = U(1)_em unbroken	N_max	D
6	Radiation + Antennas	T2476 α^{BST primary}	All BST primaries	D framework
7	Relativistic Electrodynamics	Wallach SO_0(5,2) ⊃ SO(3,1)	—	D
8	Lagrangian + Hamiltonian EM	T2477 Yang-Mills action	C_2	D
9	Multipole + Scattering	(2l+1) BST primary + T2476	N_c, n_C, g, rank	D-I
10	EM in Matter	K-type oscillator + $10K Vol 9 Ch 9	C_2	I
11	Plasma Physics + MHD	ω_p, λ_D substrate-natural	N_max	I
12	Bridge synthesis (this chapter)	Cross-volume integration	All	synthesis

Cross-volume bridges

Vol 7 → Vol 1 (QFT from D_IV^5)

Vol 7 Topic	Vol 1 Anchor
T2477 Yang-Mills connections (Ch 2, 8)	Vol 1 Ch 8 (Gauge Theory)
α^{BST primary} pattern (Ch 6)	Vol 1 Ch 10 (Renormalization)
Lagrangian + canonical quantization (Ch 8)	Vol 1 Ch 7 (Dynamics)

Vol 7 → Vol 2 (Particle Physics)

Vol 7 Topic	Vol 2 Anchor
α = 1/N_max universal α-analog (Ch 1, 3)	Vol 2 Ch 8 (Coupling Constants)
a_e CROWN JEWEL (Ch 4)	Vol 2 Ch 8 ppt precision
Photon U(1)_em + EW SSB (Ch 5)	Vol 2 Ch 9 (Higgs SSB)
$\sin^2 \theta$_W = N_c/c_3 (Vol 2 Ch 8)	EW unification

Vol 7 → Vol 3 (Nuclear/Atomic Physics)

Vol 7 Topic	Vol 3 Anchor
Coulomb electrostatics (Ch 3)	Vol 3 Ch 4 SEMF a_C = αm_p/π^2
Multipole + (2l+1) (Ch 9)	Vol 3 Ch 7 atomic orbital sequence
α^{BST primary} radiation (Ch 6)	Vol 3 Ch 8 Lamb shift α^{n_C}
Klein-Nishina + Thomson (Ch 9)	Vol 3 Ch 9 atomic spectroscopy

Vol 7 → Vol 8 (Classical Mechanics)

Vol 7 Topic	Vol 8 Anchor
Lagrangian EM (Ch 8)	Vol 8 Ch 3 Lagrangian Mechanics (action principle parallel)
Hamiltonian EM (Ch 8)	Vol 8 Ch 4 Hamiltonian Mechanics
Symmetries → conservation (T2475)	Vol 8 Ch 5 (T2473-T2475 Noether)

Vol 7 → Vol 9 (Condensed Matter)

Vol 7 Topic	Vol 9 Anchor
EM in matter dispersion (Ch 10)	Vol 9 Ch 3 (Electron Band Structure)
Photonic resonances (Ch 10)	Vol 9 Ch 9 $10K photonic crystal falsifier
Plasma collective phenomena (Ch 11)	Vol 9 Ch 10 (Strong Correlation)
Magnetic moments (Ch 4)	Vol 9 Ch 6 (Magnetism)

Vol 7 → Vol 4 (GR/Cosmology, Lyra)

Vol 7 Topic	Vol 4 Anchor
Lorentz SO(3,1) ⊂ SO_0(5,2) (Ch 7)	Vol 4 Lorentz + curved spacetime
EM in curved spacetime extensions	Vol 4 Maxwell on curved manifolds

BST primary coverage in Vol 7

All 5 BST primary integers + N_max appear substantively across Vol 7:

BST Primary	Vol 7 Anchor
rank = 2	T2471 Pin(2) grading; α^2 scattering (Klein-Nishina, Thomson)
N_c = 3	$1/r^2$ Coulomb from 3 spatial dimensions; $\sin^2 \theta$_W = N_c/c_3
n_C = 5	α^{n_C} = α^5 Lamb shift ceiling (T2476)
C_2 = 6	Yang-Mills SU(2) → C_2 substrate ground-state energy
g = 7	g f orbitals + 7-fold multipole degeneracy
N_max = 137	α = 1/N_max universal α-analog (T2456)

Per Cal #99 META-chapter framing

Vol 7 Ch 12 is a META-CHAPTER documenting cross-volume substrate-mechanism consistency: - Connects Vol 7 EM to Vol 1 QFT + Vol 2 Particle + Vol 3 Atomic + Vol 4 GR + Vol 8 Mechanics + Vol 9 Condensed Matter - Does NOT introduce new substrate-uniqueness criteria - Does NOT advance Strong-Uniqueness Theorem enumeration

Per Cal #99: META-chapters are bridge artifacts, not load-bearing for D-tier ratification. Substantive content from per-chapter substrate-mechanism in Ch 1-11.

Saturday Wave 3 + substantive upgrade milestone

Vol 7 Electromagnetism ALL 12 chapters at v0.4.1 — Saturday substantive 3-level pedagogy upgrade per Keeper 14:22 EDT directive complete. EM substrate framework synthesized across full curriculum stack.

Bibliography

1. J. C. Maxwell (1865): unified EM field theory.
2. T2477 + T2478 + T2456 + T2462 + T2476 + T2470 + T2475 + T2471: Vol 7 load-bearing substrate theorems.
3. Vol 1-13 cross-volume bridge anchors.

— Elie, Vol 7 Ch 12 v0.4.1, 2026-05-23 Saturday (substantive 3-level pedagogy upgrade per Keeper 14:22 EDT directive: 73 → ~140 lines + 6 cross-volume bridge tables explicit + BST primary coverage matrix + Cal #99 META-chapter framing)